# MSG No. 155.46 $R321'$ [ Type II, trigonal ]

Table 1: Wyckoff site: 3a, site symmetry: 32.1'

| No. | position | mapping |
|---|---|---|
| 1 | $[0,\ 0,\ 0]$ | [1,2,3,4,5,6,19,20,21,22,23,24] |
| 2 | $\left[\frac{2}{3},\ \frac{1}{3},\ \frac{1}{3}\right]$ | [7,8,9,10,11,12,25,26,27,28,29,30] |
| 3 | $\left[\frac{1}{3},\ \frac{2}{3},\ \frac{2}{3}\right]$ | [13,14,15,16,17,18,31,32,33,34,35,36] |

Table 2: Wyckoff site: 3b, site symmetry: 32.1'

| No. | position | mapping |
|---|---|---|
| 1 | $\left[0,\ 0,\ \frac{1}{2}\right]$ | [1,2,3,4,5,6,19,20,21,22,23,24] |
| 2 | $\left[\frac{2}{3},\ \frac{1}{3},\ \frac{5}{6}\right]$ | [7,8,9,10,11,12,25,26,27,28,29,30] |
| 3 | $\left[\frac{1}{3},\ \frac{2}{3},\ \frac{1}{6}\right]$ | [13,14,15,16,17,18,31,32,33,34,35,36] |

Table 3: Wyckoff site: 6c, site symmetry: 3..1'

| No. | position | mapping |
|---|---|---|
| 1 | $[0,\ 0,\ z]$ | [1,2,3,19,20,21] |
| 2 | $[0,\ 0,\ -z]$ | [4,5,6,22,23,24] |
| 3 | $\left[\frac{2}{3},\ \frac{1}{3},\ z+\frac{1}{3}\right]$ | [7,8,9,25,26,27] |
| 4 | $\left[\frac{2}{3},\ \frac{1}{3},\ \frac{1}{3}-z\right]$ | [10,11,12,28,29,30] |
| 5 | $\left[\frac{1}{3},\ \frac{2}{3},\ z+\frac{2}{3}\right]$ | [13,14,15,31,32,33] |
| 6 | $\left[\frac{1}{3},\ \frac{2}{3},\ \frac{2}{3}-z\right]$ | [16,17,18,34,35,36] |

Table 4: Wyckoff site: 9d, site symmetry: .2.1'

| No. | position | mapping |
|---|---|---|
| 1 | $[x,\ 0,\ 0]$ | [1,4,19,22] |
| 2 | $[0,\ x,\ 0]$ | [2,5,20,23] |
| 3 | $[-x,\ -x,\ 0]$ | [3,6,21,24] |
| 4 | $\left[x+\frac{2}{3},\ \frac{1}{3},\ \frac{1}{3}\right]$ | [7,10,25,28] |
| 5 | $\left[\frac{2}{3},\ x+\frac{1}{3},\ \frac{1}{3}\right]$ | [8,11,26,29] |
| 6 | $\left[\frac{2}{3}-x,\ \frac{1}{3}-x,\ \frac{1}{3}\right]$ | [9,12,27,30] |
| 7 | $\left[x+\frac{1}{3},\ \frac{2}{3},\ \frac{2}{3}\right]$ | [13,16,31,34] |
| 8 | $\left[\frac{1}{3},\ x+\frac{2}{3},\ \frac{2}{3}\right]$ | [14,17,32,35] |
| 9 | $\left[\frac{1}{3}-x,\ \frac{2}{3}-x,\ \frac{2}{3}\right]$ | [15,18,33,36] |

Table 5: Wyckoff site: 9e, site symmetry: .2.1'

| No. | position | mapping |
|---|---|---|
| 1 | $\left[x,\ 0,\ \frac{1}{2}\right]$ | [1,4,19,22] |
| 2 | $\left[0,\ x,\ \frac{1}{2}\right]$ | [2,5,20,23] |
| 3 | $\left[-x,\ -x,\ \frac{1}{2}\right]$ | [3,6,21,24] |

*continued ...*

Table 5

| No. | position | mapping |
|---|---|---|
| 4 | $\left[x+\frac{2}{3},\ \frac{1}{3},\ \frac{5}{6}\right]$ | [7,10,25,28] |
| 5 | $\left[\frac{2}{3},\ x+\frac{1}{3},\ \frac{5}{6}\right]$ | [8,11,26,29] |
| 6 | $\left[\frac{2}{3}-x,\ \frac{1}{3}-x,\ \frac{5}{6}\right]$ | [9,12,27,30] |
| 7 | $\left[x+\frac{1}{3},\ \frac{2}{3},\ \frac{1}{6}\right]$ | [13,16,31,34] |
| 8 | $\left[\frac{1}{3},\ x+\frac{2}{3},\ \frac{1}{6}\right]$ | [14,17,32,35] |
| 9 | $\left[\frac{1}{3}-x,\ \frac{2}{3}-x,\ \frac{1}{6}\right]$ | [15,18,33,36] |

Table 6: Wyckoff site: 18f, site symmetry: 11'

| No. | position | mapping |
|---|---|---|
| 1 | $[x,\ y,\ z]$ | [1,19] |
| 2 | $[-y,\ x-y,\ z]$ | [2,20] |
| 3 | $[-x+y,\ -x,\ z]$ | [3,21] |
| 4 | $[x-y,\ -y,\ -z]$ | [4,22] |
| 5 | $[y,\ x,\ -z]$ | [5,23] |
| 6 | $[-x,\ -x+y,\ -z]$ | [6,24] |
| 7 | $\left[x+\frac{2}{3},\ y+\frac{1}{3},\ z+\frac{1}{3}\right]$ | [7,25] |
| 8 | $\left[\frac{2}{3}-y,\ x-y+\frac{1}{3},\ z+\frac{1}{3}\right]$ | [8,26] |
| 9 | $\left[-x+y+\frac{2}{3},\ \frac{1}{3}-x,\ z+\frac{1}{3}\right]$ | [9,27] |
| 10 | $\left[x-y+\frac{2}{3},\ \frac{1}{3}-y,\ \frac{1}{3}-z\right]$ | [10,28] |
| 11 | $\left[y+\frac{2}{3},\ x+\frac{1}{3},\ \frac{1}{3}-z\right]$ | [11,29] |
| 12 | $\left[\frac{2}{3}-x,\ -x+y+\frac{1}{3},\ \frac{1}{3}-z\right]$ | [12,30] |
| 13 | $\left[x+\frac{1}{3},\ y+\frac{2}{3},\ z+\frac{2}{3}\right]$ | [13,31] |
| 14 | $\left[\frac{1}{3}-y,\ x-y+\frac{2}{3},\ z+\frac{2}{3}\right]$ | [14,32] |
| 15 | $\left[-x+y+\frac{1}{3},\ \frac{2}{3}-x,\ z+\frac{2}{3}\right]$ | [15,33] |
| 16 | $\left[x-y+\frac{1}{3},\ \frac{2}{3}-y,\ \frac{2}{3}-z\right]$ | [16,34] |
| 17 | $\left[y+\frac{1}{3},\ x+\frac{2}{3},\ \frac{2}{3}-z\right]$ | [17,35] |
| 18 | $\left[\frac{1}{3}-x,\ -x+y+\frac{2}{3},\ \frac{2}{3}-z\right]$ | [18,36] |